

Purification $|\Psi\rangle_{AR}$ is Unique

- up to an isometry in H_R .

Claim:

The purification vector $|\Psi\rangle_{AR}$ of a density matrix ρ_A is essentially unique. The only difference btwn purifications is an isometry (a unitary transformation for $\dim R = \text{rank } \rho_A$) on H_R .

Why?

① We constructed a purifⁿ of ρ_A over $H_A \otimes H_R$ as:

$$|\Psi\rangle_{AR} = \sum_i \sqrt{\lambda_i} |u_i\rangle_A \otimes |z_i\rangle_R$$

↳ any ONB of H_R .
 ↳ from the (unique) eigendecomⁿ of ρ_A
 $\Rightarrow$ is fixed for a given ρ_A .

② Note all purifications have $\sqrt{\lambda_i} |u_i\rangle_A$ (λ_i : eigenvalues, $|u_i\rangle$ eigenvectors) as the left tensor factor.

$\Rightarrow$ comes from the constraint that $\text{tr}_R |\Psi\rangle\langle\Psi|$ must equal ρ_A .

- take the Schmidt Decomⁿ of a pure $|\Psi_{AR}\rangle \in H_A \otimes H_B$:

$$|\Psi\rangle_{AR} = \sum_k \sqrt{c_k} |\alpha_k\rangle \otimes |\beta_k\rangle \quad \text{with } |\alpha_k\rangle, |\beta_k\rangle \text{ ONB of } H_A \text{ \& } H_R.$$

- if we add the constraint that

$$\text{tr}_R |\Psi\rangle\langle\Psi| = \rho_A,$$

then the $\{c_k, |\alpha_k\rangle\}$ are the unique eigen-decompⁿ of ρ_A .

$$\Rightarrow \text{tr}_R |\psi\rangle\langle\psi| = \text{tr}_R \left(\sum_k \sqrt{c_k} |\alpha_k\rangle \otimes |\beta_k\rangle \cdot \sum_\ell \sqrt{c_\ell} \alpha_\ell \otimes \beta_\ell \right) \quad (2)$$

$$= \text{tr}_R \sum_{k\ell} \sqrt{c_k c_\ell} |\alpha_k\rangle\langle\alpha_\ell| \otimes |\beta_k\rangle\langle\beta_\ell|$$

$$= (\mathbb{I}_A \otimes \text{tr}) \sum_{k\ell} \sqrt{c_k c_\ell} |\alpha_k\rangle\langle\alpha_\ell| \otimes |\beta_k\rangle\langle\beta_\ell|$$

$$= \sum_{k\ell} \sqrt{c_k c_\ell} |\alpha_k\rangle\langle\alpha_\ell| \otimes \text{tr} |\beta_k\rangle\langle\beta_\ell|$$

$$= \sum_{k\ell} \sqrt{c_k c_\ell} |\alpha_k\rangle\langle\alpha_\ell| \cdot \underbrace{\langle\beta_\ell|\beta_k\rangle}_{=\delta_{\ell k}}$$

$$\boxed{\text{tr}_R |\psi\rangle\langle\psi| = \sum_k c_k |\alpha_k\rangle\langle\alpha_k| \equiv \rho_A}$$

- but the $|\alpha_k\rangle$ form an ONB of H_A and the eigenvectors are the unique ONB that decompose ρ_A .

$\therefore \{c_k, |\alpha_k\rangle\}$ are the eigen-decompⁿ of ρ_A .

③ So every purification of ρ_A is in the form:

$$\boxed{|\psi\rangle_{AR} = \sum_i \sqrt{\lambda_i} |u_i\rangle \otimes |i\rangle_R \in H_A \otimes H_R}$$

$\rho_A \rightarrow$ determines λ_i & $|u_i\rangle$ and are fixed for given ρ_A .

$|i\rangle_R \rightarrow$ is any chosen ONB for H_R .

④ It's the freedom of choice for the $\{|i\rangle_R\}$ that gives the "unique up to ..." statement

$\uparrow$ choice of ONB in H_R .

$\uparrow$ fixed $\{\lambda_i, |u_i\rangle\}$ for a given ρ_A .

⑤ Suppose we have two purifications of ρ_A :

③

$$|\psi\rangle_{AR} = \sum_{k=1}^r \sqrt{\lambda_k} |u_k\rangle_A \otimes |r_k\rangle_R$$

$$|\phi\rangle_{AR'} = \sum_{k=1}^r \sqrt{\lambda_k} |u_k\rangle_A \otimes |r'_k\rangle_{R'}$$

where H_R & $H_{R'}$ are two Hilbert spaces with $\dim^F r$.

⑥ We have two ONB: $|r_k\rangle_R$
 $|r'_k\rangle_{R'}$

⑦ Define $V: H_R \rightarrow H_{R'}$ by taking ONB to ONB via.
 $|r_k\rangle \mapsto V|r_k\rangle \equiv |r'_k\rangle, k=1 \dots r$

and extend linearly from the basis to any vector.

⑧ V is necessarily unitary $\because$ the $|r_k\rangle, |r'_k\rangle$ are ONB!

Check: $V^\dagger V = I$

- we have $\langle r'_k | r'_k \rangle = \langle r_k | V^\dagger V | r_k \rangle$ by ⑦.

- but lhs = δ_{kk} .

- $\therefore$ rhs $\Rightarrow \langle r_k | V^\dagger V | r_k \rangle = \delta_{kk}$.

$[V^\dagger V]_{kk} = \delta_{kk}$ as matrix elements.

$$\therefore \boxed{V^\dagger V = I}$$

- swap $r' \leftrightarrow r$ above to get $\boxed{VV^\dagger = I}$

- $\therefore V$ is unitary.

⑨ Define operator $I_A \otimes V$ and operate on $|\psi\rangle$:

$$(I_A \otimes V) |\psi\rangle_{AR} = \sum_k \sqrt{\lambda_k} (I_A \otimes V) \cdot (|u_k\rangle \otimes |r_k\rangle_R)$$

$$= \sum_k \sqrt{\lambda_k} |u_k\rangle \otimes V|r_k\rangle_R$$

$$= \sum_k \sqrt{\lambda_k} |u_k\rangle \otimes |r'_k\rangle_{R'} = |\phi\rangle_{AR'}$$

$$\boxed{|\phi\rangle_{AR'} = (I_A \otimes V) |\psi\rangle_{AR}}$$

⑩ So all purifications of a given ρ are related ④
 via:

$$|\phi\rangle_{AR'} = (I_A \otimes V) |\psi\rangle_{AR}$$

for any unitary $V: H_R \rightarrow H_{R'}$, both dim rank (ρ) .

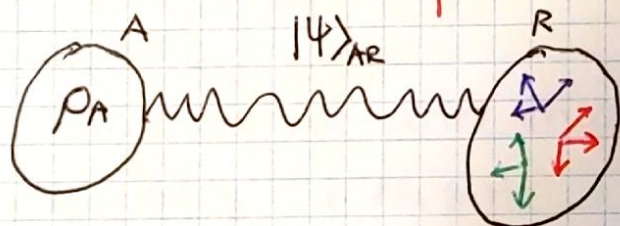
→ if $\dim R \neq \dim R' \geq \text{rank}(\rho)$ then it's an isometry.

HJW Theorem:

A practical application of purification.

Part 1:

Given some purification of ρ_A to a $|\psi\rangle_{AR}$, any measurement in R produces an ensemble in A for ρ_A .



Measure in R



Ensemble in A

$$\{|\phi_i\rangle_A, p_i\}$$

$$\{|\chi_i\rangle_A, q_i\}$$

$$\{|\xi_i\rangle_A, r_i\}$$

• choosing a measurement basis in R allows us to tune in different ensembles for a given ρ_A in a controlled manner.

Warm-Up.

⑤

- Start with the eigendecomposition of a d.m. ρ_A in H_A .

$$\rho_A = \sum_{i=1}^r \lambda_i |e_i\rangle\langle e_i|$$

- pick $r=2$ for definiteness and say we do a purification:

$$|\psi\rangle = \alpha_0 |e_0\rangle_A |0\rangle_R + \alpha_1 |e_1\rangle_A |1\rangle_R.$$

- now measure R in the computational basis: $\{|0\rangle_R\langle 0|_R, |1\rangle_R\langle 1|_R\}$

- we leave A alone.

- given outcome $|0\rangle_R$, the resultant state is the projection:

$$(I_A \otimes |0\rangle_R\langle 0|_R) |\psi\rangle.$$

$$\Rightarrow (I_A \otimes |0\rangle\langle 0|) \cdot (\alpha_0 |e_0\rangle |0\rangle + \alpha_1 |e_1\rangle |1\rangle).$$

$$= \alpha_0 |e_0\rangle \otimes |0\rangle\langle 0|_R + \alpha_1 |e_1\rangle \otimes |0\rangle\langle 1|_R.$$

$$= \alpha_0 |e_0\rangle_A \otimes |0\rangle_R$$

- so the state on AR after measurement outcome $|0\rangle_R$ on R is:

$$\alpha_0 |e_0\rangle_A \otimes |0\rangle_R \in H_A \otimes H_R \quad (*)$$

- discarding R we have: $\alpha_0 |e_0\rangle_A \in H_A$

- we do this by multiplying $(*)$ by $I \otimes \langle 0|_R$:

$$(I \otimes \langle 0|) \cdot (\alpha_0 |e_0\rangle \otimes |0\rangle) = \alpha_0 |e_0\rangle \otimes \langle 0|0\rangle = \alpha_0 |e_0\rangle.$$

- so together: ① projective measurement in R on AR system
+ ② discard R .

$$\Rightarrow (I \otimes \langle 0|) \cdot (I \otimes |0\rangle\langle 0|) \cdot |\psi\rangle$$

$$= I_A \otimes \langle 0|_R.$$

- for a general projection in R on a basis vector $|\eta_i\rangle$ we have:

$$(I \otimes \langle \eta_i|) \cdot (I \otimes |\eta_i\rangle\langle \eta_i|) \cdot |\psi\rangle_{AR} = (I \otimes \langle \eta_i|) |\psi\rangle_{AR} \in H_A.$$

Back to Proof - measuring in R on a purification of ρ_A induces an ensemble in A . ⑥

- start again with eigendecomps of ρ_A :

$$\rho_A = \sum_{i=1}^r \lambda_i |e_i\rangle\langle e_i|.$$

- purify it by choosing a basis in a reference system R :

$$|\psi\rangle_{AR} = \sum_i \sqrt{\lambda_i} |e_i\rangle_A \otimes |i\rangle_R. \quad \rightarrow \text{from } X \equiv \sum_i \sqrt{\lambda_i} |e_i\rangle_A \langle i|_R \rightarrow \text{vec } X.$$

- now pick an ONB $|\eta_i\rangle \in H_R$ for projective measurements in R
 $|\eta_i\rangle\langle\eta_i| \in L(H_R), \quad i=1 \dots r.$

- extend to a measurement on the joint system AR , leaving A alone.

$$I_A \otimes |\eta_i\rangle\langle\eta_i| \in L(H_A \otimes H_R), \quad i=1 \dots r$$

- from the warm-up we know the post-measurement state in H_A conditioned on outcome $|\eta_i\rangle$ is:

$$(I_A \otimes \langle\eta_i|) |\psi\rangle_{AR} = (I_A \otimes \langle\eta_i|) \cdot \sum_{j=1}^r \sqrt{\lambda_j} |e_j\rangle_A \otimes |j\rangle_R$$

$$= \sum_j \sqrt{\lambda_j} |e_j\rangle \cdot \langle\eta_i|j\rangle_R \in H_A.$$

$$= \sum_j \sqrt{\lambda_j} |e_j\rangle \cdot \langle j|\eta_i\rangle^*$$

$$= \sum_j \sqrt{\lambda_j} |e_j\rangle \eta_i^{j*}$$

$\leftarrow j$ 'th coord of $|\eta_i\rangle$ in H_R .

- note that: $X|j\rangle_R = \sum_i \sqrt{\lambda_i} |e_i\rangle_A \langle i|_R \cdot |j\rangle_R = \sqrt{\lambda_j} |e_j\rangle_A$

- So, the post-measurement state in A given outcome $|\eta_i\rangle$ in R is:

$$\begin{aligned} (I_A \otimes \langle \eta_i |) |\Psi\rangle_{AR} &= \sum_j X |j\rangle_R \eta_i^{j*} \\ &= X \sum_j \eta_i^{j*} |j\rangle \end{aligned}$$

discarded
the

(*) $(I_A \otimes \langle \eta_i |) |\Psi\rangle_{AR} \equiv X |\eta_i^*\rangle_R$ (is not normalized yet).

- note we've defined $|\eta_i^*\rangle$ as the same as $|\eta_i\rangle$ except its components are conjugated.

- define the unnormalized post-measurement state in H_A as:

$$|\tilde{\chi}_i\rangle \equiv X |\eta_i^*\rangle_R$$

- the probability of measurement outcome i is:

$$p_i = \langle \Psi | I_A \otimes |\eta_i\rangle \langle \eta_i| | \Psi \rangle.$$

$$= \langle \Psi | (I \otimes |\eta_i\rangle) \circ (I \otimes \langle \eta_i|) | \Psi \rangle, \quad \leftarrow \text{factor projection.}$$

$$p_i = \langle \tilde{\chi}_i | \tilde{\chi}_i \rangle \quad \leftarrow \text{from (*) above.}$$

- so the normalized outcome in H_A is:

$$|\phi_i\rangle = \frac{|\tilde{\chi}_i\rangle}{\sqrt{\langle \tilde{\chi}_i | \tilde{\chi}_i \rangle}} = \frac{|\tilde{\chi}_i\rangle}{\sqrt{p_i}}$$

- note that treating the outcomes in A as an ensemble gives us the density matrix:

⑧

$$\sum_i p_i |\phi_i\rangle \langle \phi_i|$$

$$= \sum_i p_i \frac{1}{(\sqrt{p_i})^2} X |\eta_i^*\rangle \langle \eta_i^*| X^\dagger$$

$$= X \left(\sum_i |\eta_i^*\rangle \langle \eta_i^*| \right) X^\dagger$$

$$= X X^\dagger$$

$$= \rho_A.$$

- so the $\{|\phi_i\rangle, p_i\}$ are an ensemble for ρ_A !

- this means any projective ONB measurement on the reference system with a purification state of ρ_A produces a pure state ensemble for ρ_A .

DONE

QUESTIONS:

- ① can all ensembles be generated this way?
- ② if so, can we fix a purification state $|\Psi\rangle_{AR}$ and then generate all ensembles by tuning measurements?

⇒ THIS IS THE CONVERSE STATEMENT.

Answer: ① YES!

② YES!

Bonus Question:

What ensemble in A is generated if we choose the purification basis $|i\rangle_R$ for measurements → Ans: What ensemble do we know?